

CMP5354 Software Design

Physical Asset Tracking Vision Platform

(BMW Hams Hall Plant)

D2 Software Solution

Abraham Sharkey

Computer Science

Birmingham City University

Birmingham, United Kingdom

abmsharkey@gmail.com

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System Overview

The proposed solution is an indoor Vehicle Tracking Platform deployed within the BMW Hams Hall plant. The system uses the existing CCTV camera infrastructure together with computer vision techniques to continuously detect, identify, and track logistics vehicles in real time. Each vehicle is distinguished using a unique fiducial marker identifier (AprilTag), enabling reliable differentiation during operation.

The platform maintains a continuously updated tracking snapshot containing the latest X, Y, and Z position of each detected vehicle relative to the facility layout. This allows the Logistics Control Operator to monitor live vehicle positions through the tracking interface. In addition, the system can raise maintenance-related alerts when a tracked vehicle requires attention and can identify non-vehicle obstructions that may affect tracking continuity or path decisions.

To support wider system integration, the platform provides vehicle tracking data to BMW's Digital Twin System through a secure authenticated API. Administrative tasks, including tag-to-vehicle mapping management and CCTV status monitoring, are handled by the IT Engineer to maintain identification accuracy and infrastructure reliability. Access to tracking data is restricted to authorised systems only, supporting the organisation's security and data protection requirements.

Functional Requirements (FRs)

The functional requirements in Table 1 define the core capabilities of the proposed BMW logistics vehicle tracking system. FR1-FR4 are the baseline requirements carried forward from D1, while FR5-FR7 are additional requirements introduced in D2. Related sub-requirements have been included to improve clarity and traceability across the later design models.

FR ID	Description	Functional Requirement
FR1	Vehicle detection and tracking	The system must detect and track the actual logistics vehicles in real-time through the existing CCTV video feeds.
FR1.1	Multi-vehicle detection	The system must detect multiple logistics vehicles within the same CCTV frame where they are visible at the same time.
FR1.2	Continuous tracking update	The system must continuously update the tracked position of each detected logistics vehicle while it remains within camera coverage.
FR2	Vehicle positioning	The system must be able to depict the location of each individual logistics vehicle in relation to both the actual area of the facility and its coordinates.
FR2.1	Layout-relative positioning	The system must display each tracked logistics vehicle relative to the facility layout or monitored plant area.
FR2.2	Coordinate generation	The system must generate coordinate values for each tracked logistics vehicle to represent its current position.
FR3	Maintenance alerting	The system should be able to raise the operating status of each logistics vehicle through a system alert for any necessary maintenance to be applied.
FR3.1	Maintenance alert creation	The system should generate an alert when a logistics vehicle is identified as requiring maintenance attention.
FR3.2	Alert information display	The system should present the relevant vehicle identifier and operating status information within the alert.

FR4	Obstruction awareness	The system must regard any other obstructions besides the logistics vehicles within the facility that can potentially affect the journey to improve tracking and influence path decisions.
FR4.1	Obstruction detection	The system must detect non-vehicle obstructions present within the monitored facility space.
FR4.2	Obstruction impact flagging	The system must flag when a detected obstruction may affect vehicle tracking continuity or path decisions.
FR5	Vehicle differentiation	The system must distinguish between individual logistics vehicles during tracking by using a unique fiducial marker identifier.
FR5.1	Marker recognition	The system must recognise and decode the fiducial marker attached to each logistics vehicle.
FR5.2	Identity assignment	The system must assign a unique tracked identity to each detected logistics vehicle during operation.
FR6	CCTV status monitoring	The system must continuously monitor the operational status of all connected CCTV cameras.
FR6.1	Offline camera detection	The system must flag a connected camera as offline if it does not provide a frame within a defined threshold period.
FR6.2	Technical fault notification	The system must notify the IT Engineer when a connected camera becomes unavailable.
FR7	Digital Twin API integration	The system must provide a secure API through which BMW's Digital Twin can request vehicle tracking data.
FR7.1	Authenticated access	The system must restrict API access to authenticated requests only.
FR7.2	Structured data output	The system must return vehicle tracking data in a standardised structured format suitable for Digital Twin integration.

Table 1. Functional Requirements

Non-Functional Requirements (NFRs)

The non-functional requirements in Table 2 define the measurable quality constraints that govern how the proposed BMW vehicle tracking platform must perform. They focus on performance, identification reliability, and security because these qualities are the most critical to the system's real-time tracking, vehicle differentiation, and Digital Twin integration.

NFR ID	Description	Non-Functional Requirement
NFR1	Performance under concurrent load	The system must provide vehicle position updates via the API within 1 second of capturing a camera frame while handling up to 50 concurrent API requests.
NFR2	Identification reliability	The system must successfully decode and classify fiducial marker identifiers with at least 95% accuracy within a calibrated operating range of 0.5 to 5 metres under standard factory lighting conditions.
NFR3	Data security	The system must reject 100% of API requests that do not contain a valid authentication token.

Table 2. Non-Functional Requirements

Ethical & Legal Issues

The issues in Table 3 identify the main ethical and legal concerns associated with deploying the proposed BMW tracking platform in a live manufacturing environment. The focus here is on why these issues are problematic for the system and its use context, rather than how they would be mitigated.

ID	Issue	Justification	Related FR/NFR
E1	Indirect employee surveillance	Continuous CCTV-based vehicle tracking may indirectly reveal worker movement patterns, routines, or behaviours even when the system is intended to monitor vehicles rather than people. In a workplace setting, this creates ethical concerns around proportionality, transparency, and staff awareness.	FR1, FR5, FR7, NFR3
L1	UK GDPR and indirect identifiability	Under Article 4(1) of Regulation (EU) 2016/679, personal data includes information relating to an identified or identifiable natural person. If vehicle location data, marker identities, and shift or access records are combined, operators may become indirectly identifiable, bringing the system within the scope of data protection obligations (European Parliament and Council, 2016).	FR5, FR7, NFR3
L2	Liability from inaccurate location or identification data	If the system produces incorrect vehicle positions or misidentifies a tracked vehicle, operational decisions may be made using inaccurate information. In a live industrial environment, this could contribute to unsafe movement, workflow disruption, or collisions, creating potential legal and organisational liability under workplace health and safety duties (Health and Safety at Work etc. Act 1974).	FR2, FR5, NFR2
E2	Uneven system accuracy and unfair operator inference	If tracking accuracy varies across different areas of the plant because of lighting, camera angle, or partial occlusion, some vehicles or operators may be represented less accurately than others. If those outputs are later used to infer efficiency, compliance, or behaviour, this may lead to unfair judgements and reduced trust in the system.	FR1, FR5, FR6, NFR2
L3	HSE compliance in safety-critical use	If BMW later relies on this platform for safety-critical purposes such as collision avoidance, movement control, or hazard prevention, incorrect or delayed outputs could create compliance issues. A tracking platform used beyond its validated purpose may fail to satisfy the legal duties imposed by workplace safety law (Health and Safety at Work etc. Act 1974; The Management of Health and Safety at Work Regulations 1999).	FR2, FR4, NFR1, NFR2

Table 3. Ethical & Legal Issues

Risk Assessment

The risks in Table 4 identify the main technical and operational failure scenarios that could affect the reliability, performance, and safe deployment of the proposed BMW vehicle tracking platform. Each risk is linked to the relevant functional or non-functional requirements and paired with a targeted mitigation to preserve system continuity in a live factory environment.

Id	Risk	Severity	Mitigation	Related FR/NFR
Ri1	Occlusion or lighting variation causes temporary loss of vehicle tracking.	Serious	Implement predictive tracking using Kalman filtering during short visibility loss and use high-contrast fiducial markers with built-in error correction to improve detection robustness under variable lighting and partial occlusion.	FR1, FR1.2, FR5, NFR2
Ri2	Marker aliasing causes incorrect vehicle identification.	Serious	Enforce a minimum Hamming distance between marker IDs and apply temporal consistency checks to reject implausible identity switches between frames.	FR5, FR5.1, FR5.2, NFR2
Ri3	Network latency causes API updates to miss the real-time performance target.	Moderate	Use edge processing so video frames are analysed locally, and only structured tracking data is transmitted to the API layer rather than full video streams.	FR2, FR7, NFR1
Ri4	CCTV camera failure or network disconnection interrupts active monitoring.	Serious	Continuously monitor camera heartbeats and flag a camera as offline if no frame is received within the defined threshold, then send an automated alert to the IT Engineer.	FR6, FR6.1, FR6.2
Ri5	Excessive concurrent API requests from the Digital Twin overload the endpoint.	Moderate	Apply API rate limiting and request throttling at the gateway to prevent request bursts from degrading the core tracking engine's response time.	FR7, FR7.1, NFR1

Table 4. Risk Assessment

Use Case Diagram (UCD)

Figure 1 presents the Use Case Diagram for the Vehicle Tracking Platform. It models the main external interactions between the system and its three actors: the Logistics Control Operator, the Digital Twin System, and the IT Engineer. The diagram covers FR1-FR7 and uses verb-based use cases together with <<include>> and <<extend>> relationships to show mandatory and conditional system behaviour.

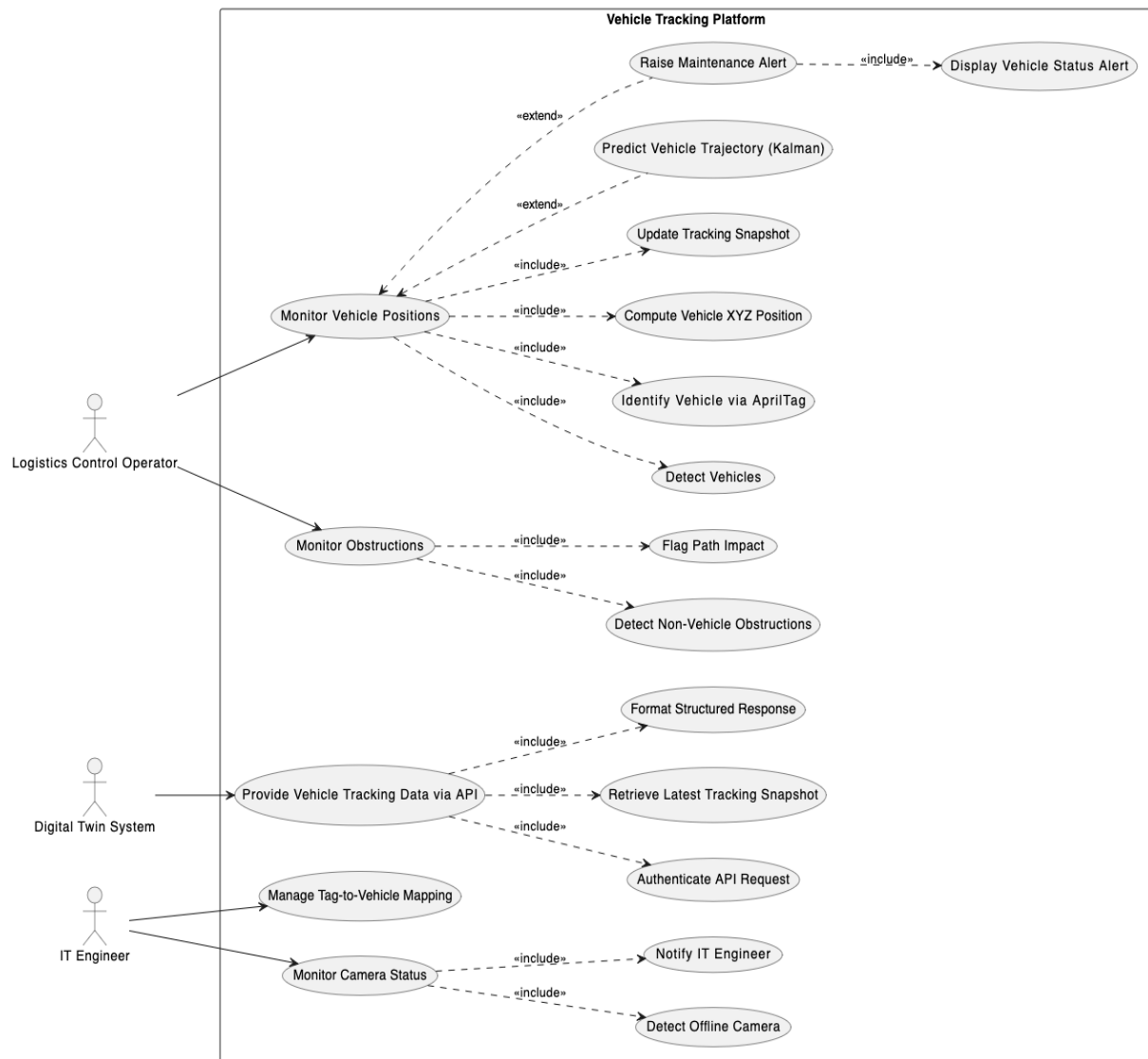


Figure 1. Use Case Diagram (UCD)

Use Case Specification (UCS)

Table 5 expands the “Monitor Vehicle Positions” use case from Figure 1. It describes the normal monitoring flow together with the main alternative flows associated with vehicle detection, AprilTag-based identification, XYZ position computation, and tracking snapshot updates.

Name	Monitor Vehicle Positions
Purpose	To enable the Logistics Control Operator to monitor the real-time XYZ positions of logistics vehicles through continuous detection, AprilTag-based identification, XYZ position computation, and tracking snapshot updating.
Prerequisites	1. CCTV camera streams are active. 2. Tag-to-vehicle mappings are configured. 3. Calibration data is available for XYZ position computation. 4. The processing pipeline is operational.
Narrative	1. The Logistics Control Operator initiates live monitoring. 2. The system captures a CCTV frame. 3. The system detects vehicles within the frame. 4. The system identifies each detected vehicle via AprilTag. 5. The system computes each vehicle’s XYZ position relative to the facility layout. 6. The system updates the tracking snapshot with the latest vehicle identity and XYZ position. 7. The system displays live vehicle positions and repeats steps 2-7 continuously while monitoring remains active.
Outcome	The tracking snapshot continuously reflects the most recent vehicle identities and XYZ positions, enabling real-time monitoring.
Alternative Narrative	Occlusion Detected (Ri1): If tag visibility is temporarily lost, the system invokes Predict Vehicle Trajectory (Kalman) to estimate an interim vehicle position.

	No Vehicles Detected (A1): If no vehicles are detected in the current frame, the system records “no detections” for that frame and continues monitoring. AprilTag Unreadable (A2): If a vehicle is detected but its AprilTag cannot be decoded, the system assigns the temporary identity “Unknown” and continues tracking the detected object. Computation Failure (A3): If XYZ position computation fails for a detected vehicle, the system logs the failure for that frame and does not commit invalid position data to the tracking snapshot.
Alternative Narrative Outcome	The system continues processing subsequent frames without interruption. Where prediction is used, interim positions are labelled “estimated” until direct visibility is restored. Where no detection or computation failure occurs for a frame, the system avoids committing invalid tracking data and resumes normal processing on the next frame.

Table 5. Use Case Specification

Class Diagram (CD)

Figure 2 presents the Class Diagram for the Vehicle Tracking Platform. It models the static structure of the solution by showing the main classes, attributes, operations, data types, and associations required to support tracking, vehicle differentiation, obstruction awareness, maintenance alerting, camera status monitoring, and authenticated API integration.

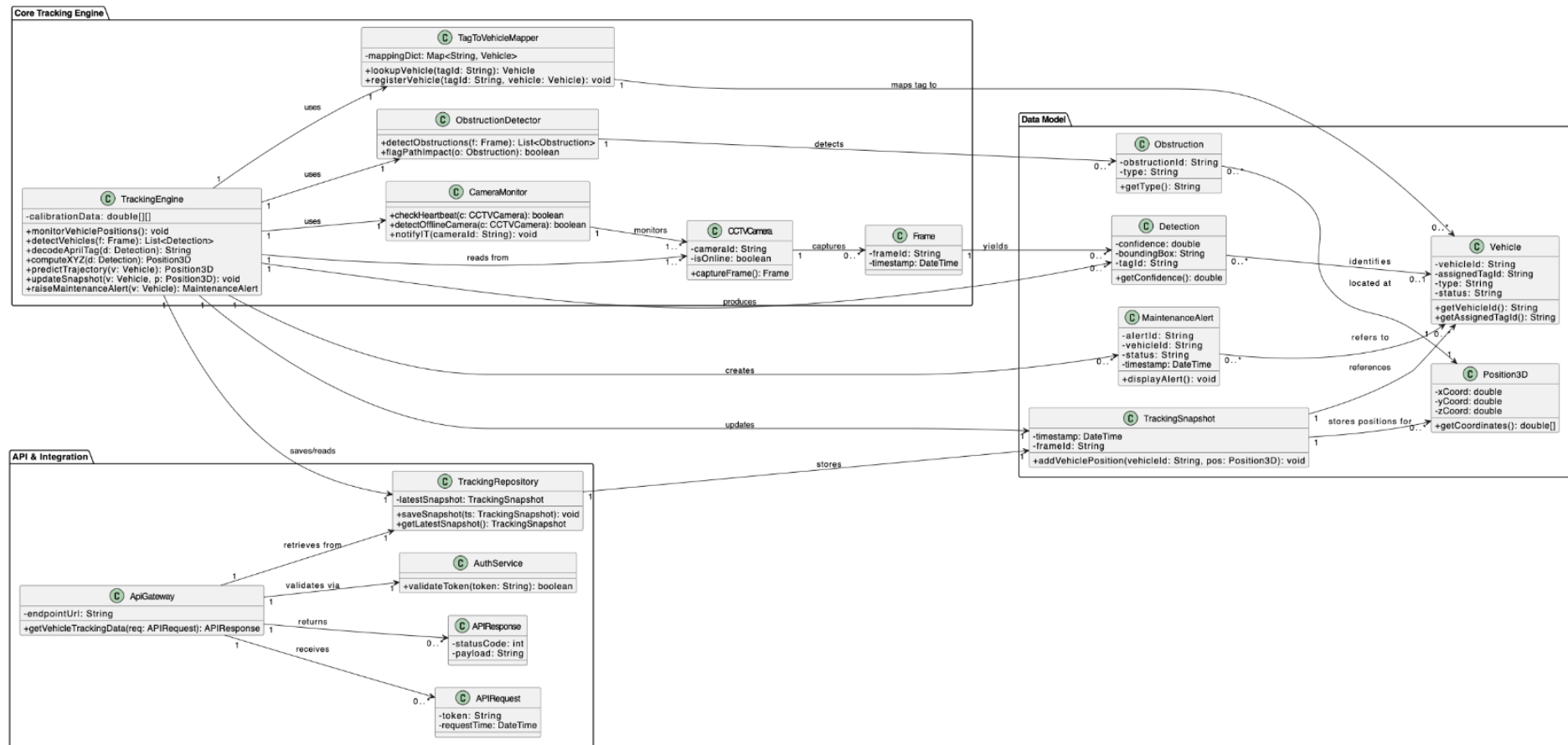


Figure 2. Class Diagram (CD)

Sequence Diagram (SD)

Figure 3 illustrates the dynamic interaction for the “Monitor Vehicle Positions” use case. The participating objects are derived from the Class Diagram, ensuring consistency between the system’s static structure and runtime behaviour. The sequence uses loop and alt fragments to represent continuous monitoring and the main alternative scenarios defined in the UCS, including occlusion, unreadable tags, no detections, and computation failure.

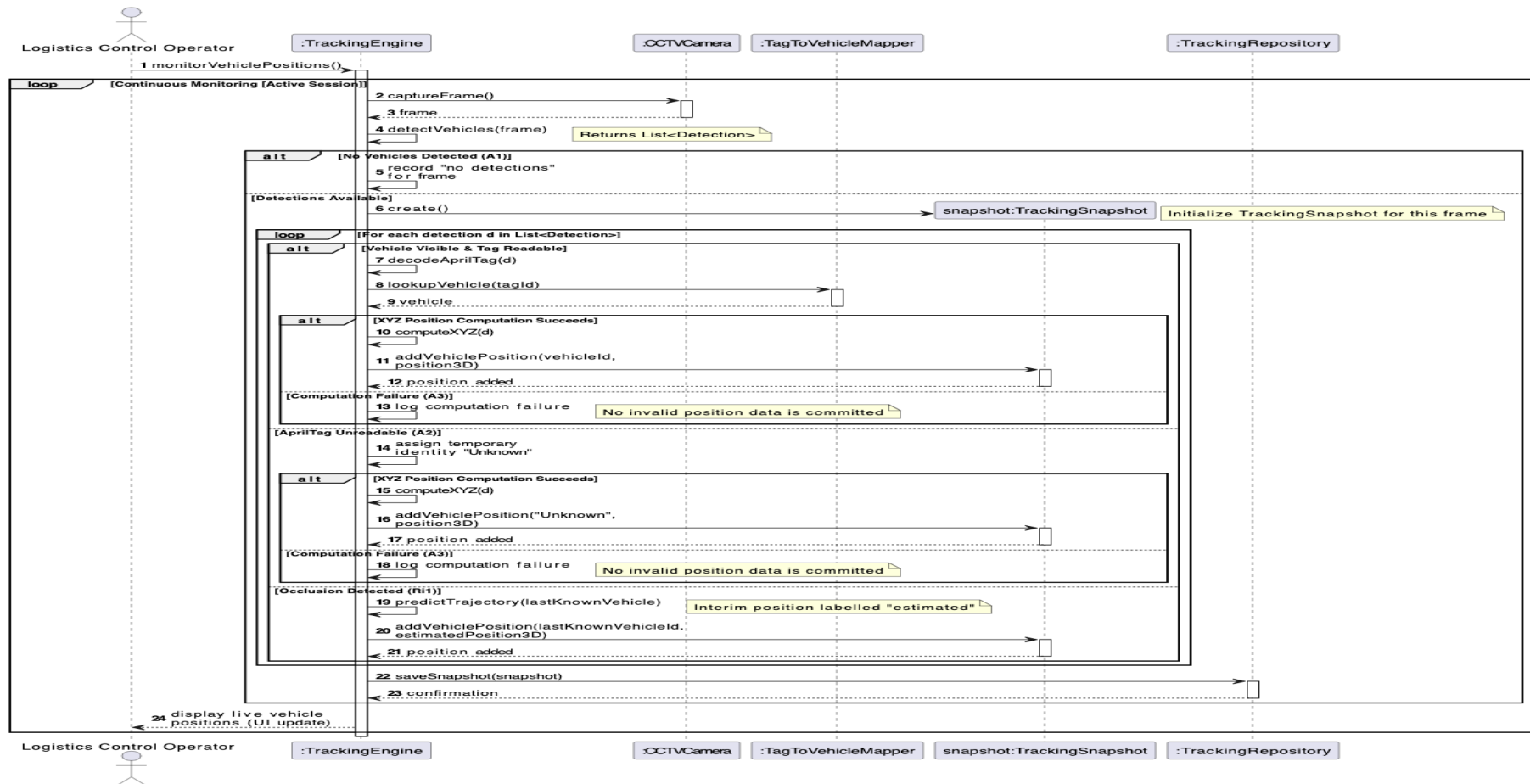


Figure 3. Sequence Diagram (SD)

References

European Parliament and Council (2016) Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation). OJ, L119, 4 May, pp. 1-88.

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